Sampling in Action: The M&M Challenge

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Roadmap

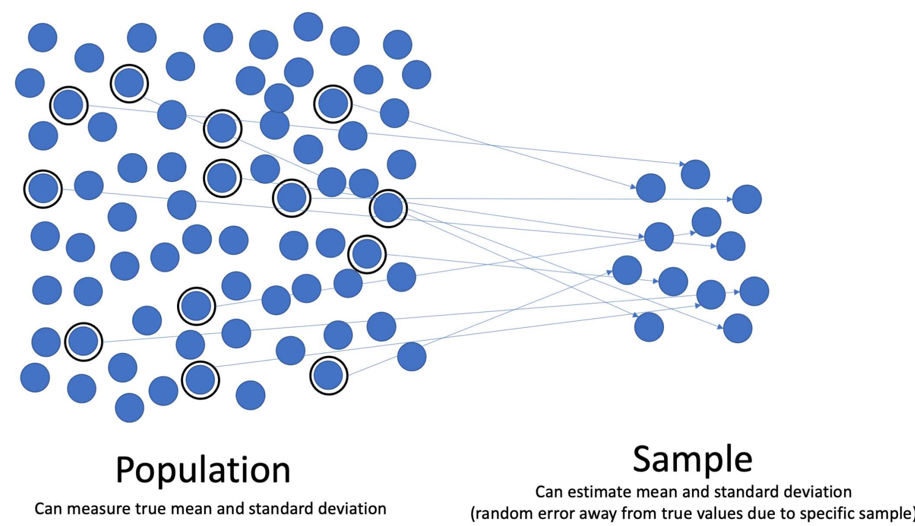
- Activity: count and enter M&M colors (15 min)
- Class analysis: live charts from your data (20 min)
- Concepts: why sample size and design matter (10 min)
- Wrap-up and takeaways (5 min)

By the end of this lecture, you should be able to:

- Distinguish between samples and populations and explain why we sample
- Explain how sample size affects the stability of estimates
- Identify sources of bias in sampling (production, selection, measurement, non-response)
- Evaluate whether a sample is representative of its target population

Quick refresher: what is sampling?

- We study a part (sample) to learn about the whole (population).
- Samples differ from each other; that's normal variability.
- Our goal today: estimate each color's percentage in the wider "population" of candies.
- We'll see that combining more bags leads to more stable class estimates.



M&M Sampling Activity

- Objective: Demonstrate sampling principles using M&M's
  - Hands-on experience with data collection and analysis
- Materials:
  - Small packages of plain M&M's (one per student)
  - Napkins for sorting
- Outcome: start with your bag, then build to a class-wide estimate

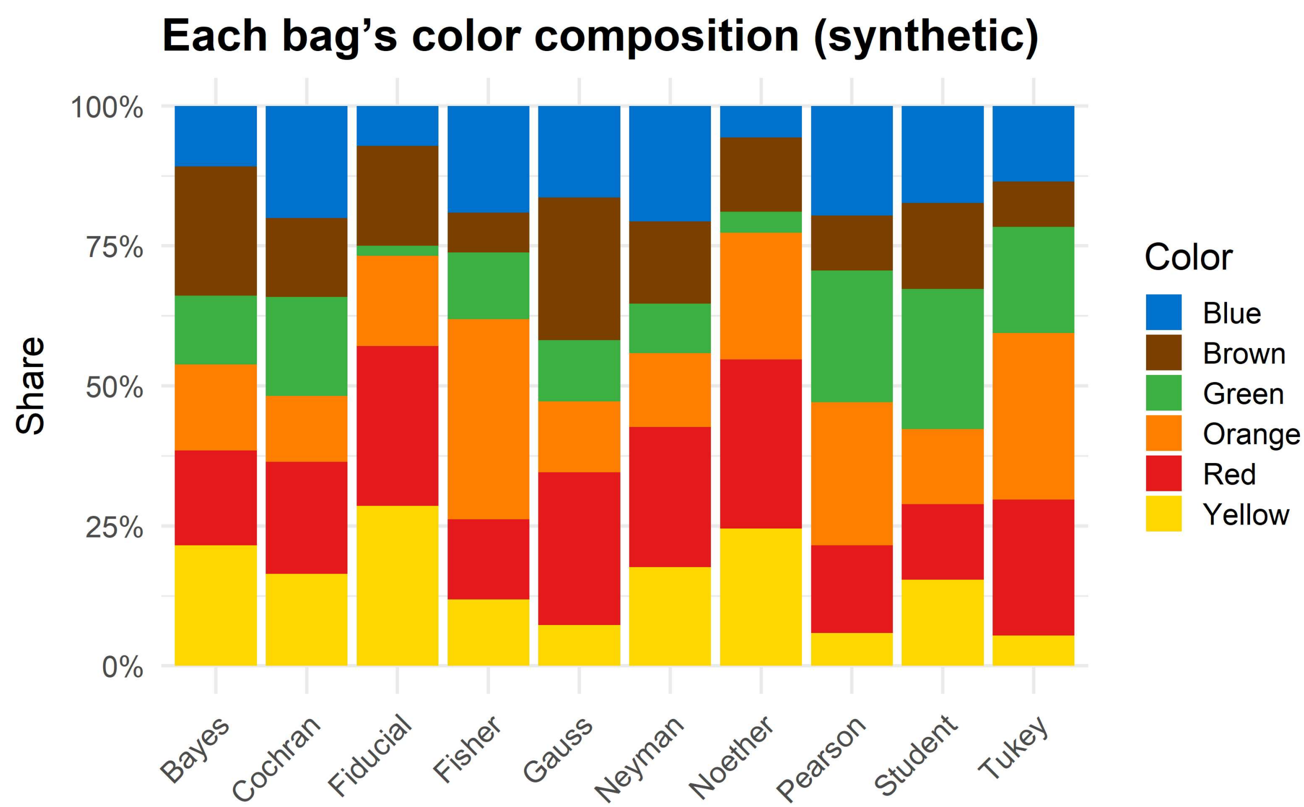
M&M Sampling Procedure

- Distribute M&M packages and materials
- Sort M&M's by color onto your napkin
- Count each color: Blue, Orange, Green, Red, Yellow, Brown.
- Enter your counts (raw numbers; 0 if a color is absent).
- Hypothesize population color distribution
- Compare with a partner: are your percentages similar?
- We’ll pool the entire class and visualize the results
  - using Google Sheets (and some R magic)

What to expect...

Analysis in Action

- What we'll get from the class data



Source Code

```
library(gridExtra)
set.seed(123) # For reproducibility

# Define the number of students and colors
students <- c("Tukey", "Gauss", "Noether", "Fisher",
              "Bayes", "Pearson", "Student",
              "Fiducial", "Neyman", "Cochran")
base_cols <- c(
  Blue   = "#0072CE",
  Orange = "#FF7F00",
  Green  = "#3CB043",
  Red    = "#E41A1C",
  Yellow = "#FFD700",
  Brown  = "#7B3F00"
)
colors <- names(base_cols)

# Simulate the total number of M&Ms for each student
bag_sizes <- sample(15:20, length(students), replace = TRUE)


# Simulate the counts of each color for each student
color_counts <- replicate(length(colors),
                          sample(1:bag_sizes,
                                 length(students), replace = TRUE))

# Create the dataframe
df_syn <- data.frame(Name = students,
                     color_counts)

colnames(df_syn)[-1] <- colors

# Calculate the percentages
df_syn <- df_syn %>%
  mutate(Total = Blue + Orange + Green + Red + Yellow + Brown)

df_long_syn <- df_syn %>%
  pivot_longer(Blue:Brown, names_to="Color", values_to="Count") %>%
  group_by(Name) %>% mutate(Prop = Count/sum(Count)) %>% ungroup()

# Plotting the data
p_syn_bags <- ggplot(df_long_syn, aes(Name, Prop, fill = Color)) +
  geom_col() +
  scale_fill_manual(values = base_cols) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Each bag’s color composition (synthetic)", x = NULL, y = "Share") +
  thm + theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Overall distribution of M&Ms
overall_distribution <- df_syn %>%
  select(Blue, Orange, Green, Red, Yellow) %>%
  summarise(across(everything(), sum)) %>%
  pivot_longer(cols = everything(), names_to = "Color", values_to = "Count")

p_syn_total <- df_syn %>%
  summarise(across(Blue:Brown, sum)) %>%
  pivot_longer(everything(), names_to = "Color", values_to = "Total") %>%
  ggplot(aes(Color, Total, fill = Color)) +
  geom_col() +
  scale_fill_manual(values = base_cols, guide = "none") +
  labs(title = "Total M&Ms by color (synthetic)", x = NULL, y = "Total") +
  thm
p_syn_bags
```

```
#gridExtra::grid.arrange(p_syn_bags, p_syn_total, ncol = 2)

# Display both plots
#grid.arrange(stacked_plot, overall_plot, ncol = 2)
```

Methods in Psychological Research

Data Collection

Scan to input your data!

Checklist:

- Each row = one bag (add your name/initials)
- Enter raw counts (not percentages)
- Include all six colors (use 0 if needed)
- Submit once (no duplicates)

Discussion prompts:

- Which colors have the widest ranges? What determines the width?
- How would doubling N change these intervals?

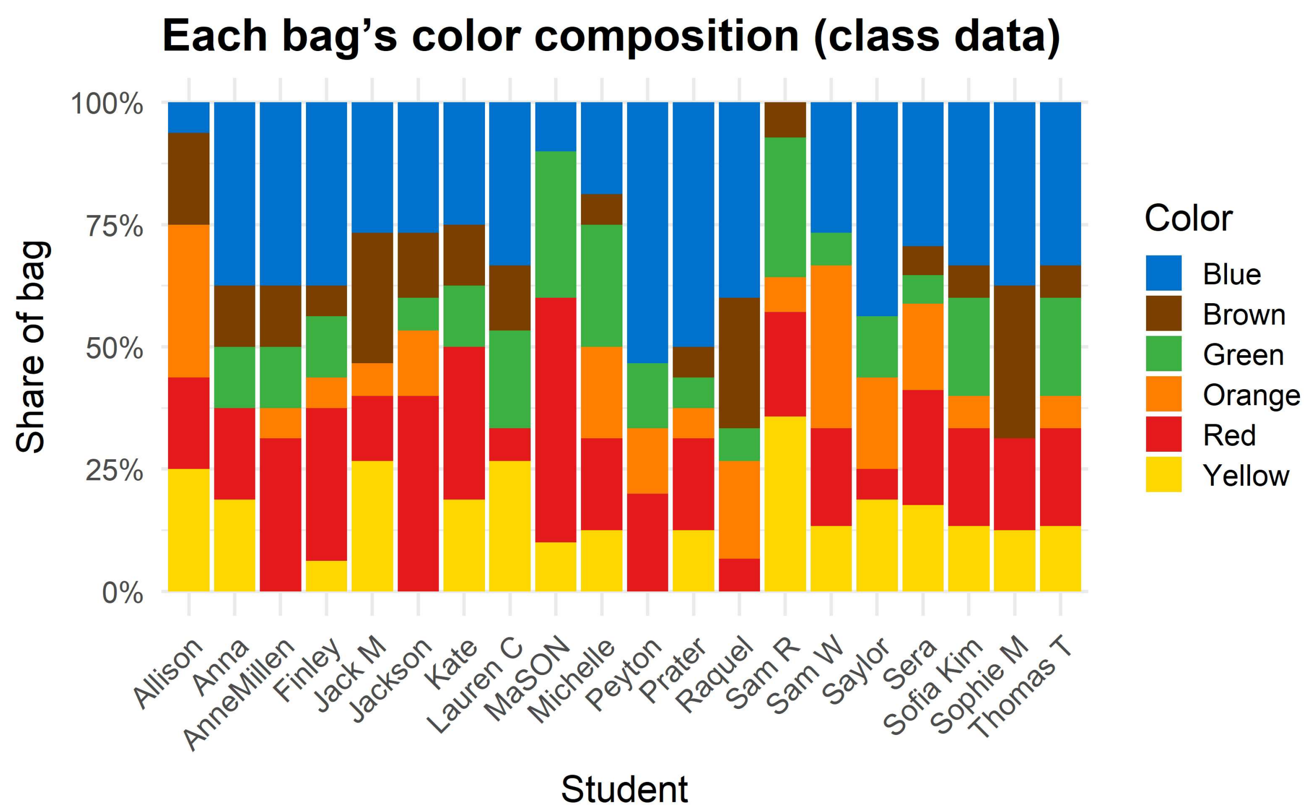
https://docs.google.com/spreadsheets/d/1D4i8e0pTrqwLk_FjMFtkimqhmtOrBf-X6OU9RT57m_Q

Analysis in action

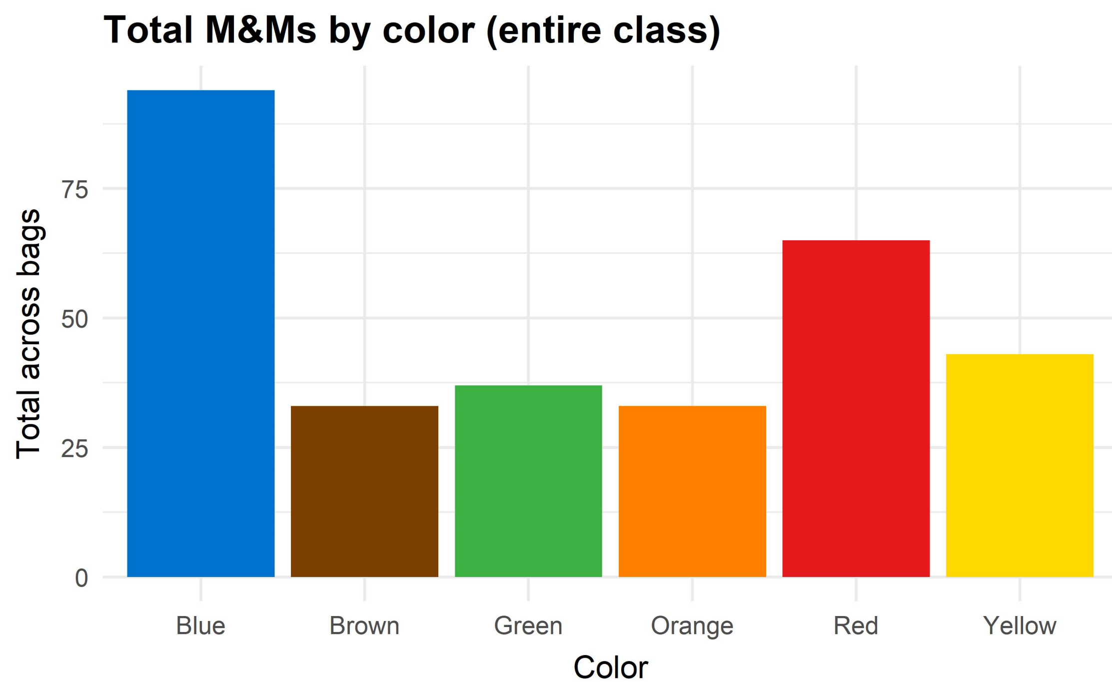
Discussion prompts:

- Which colors have the widest intervals? What determines the width?
- How would doubling N change these intervals?

What do our samples look like?



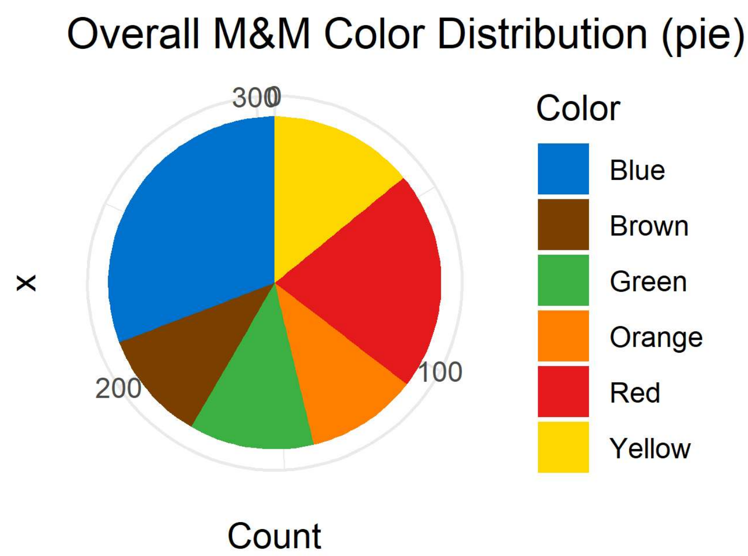
Class pooled counts

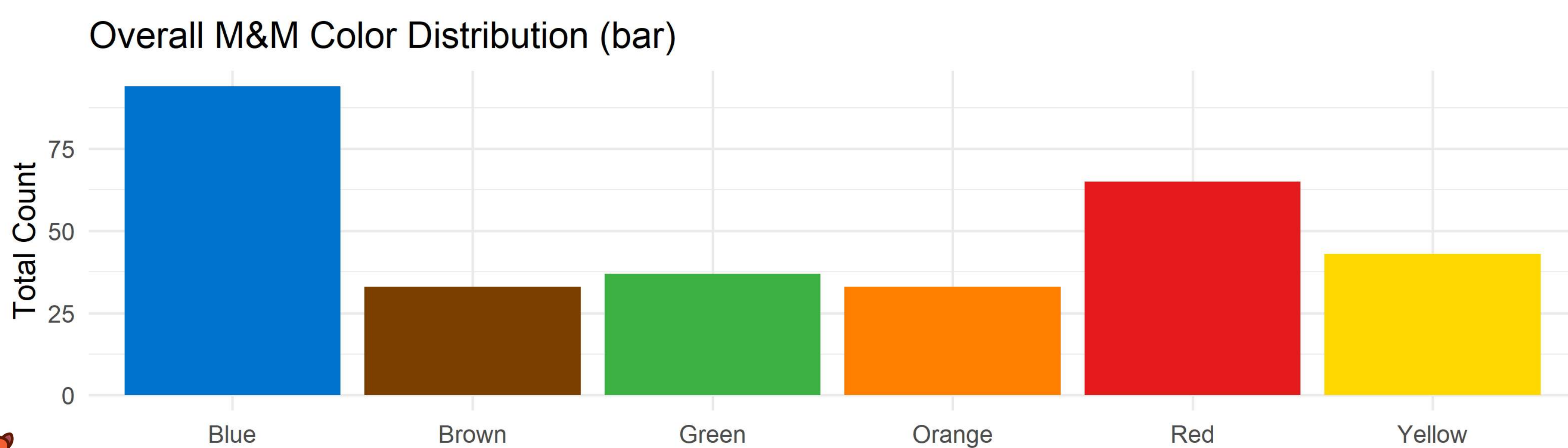


Sample Size Effects

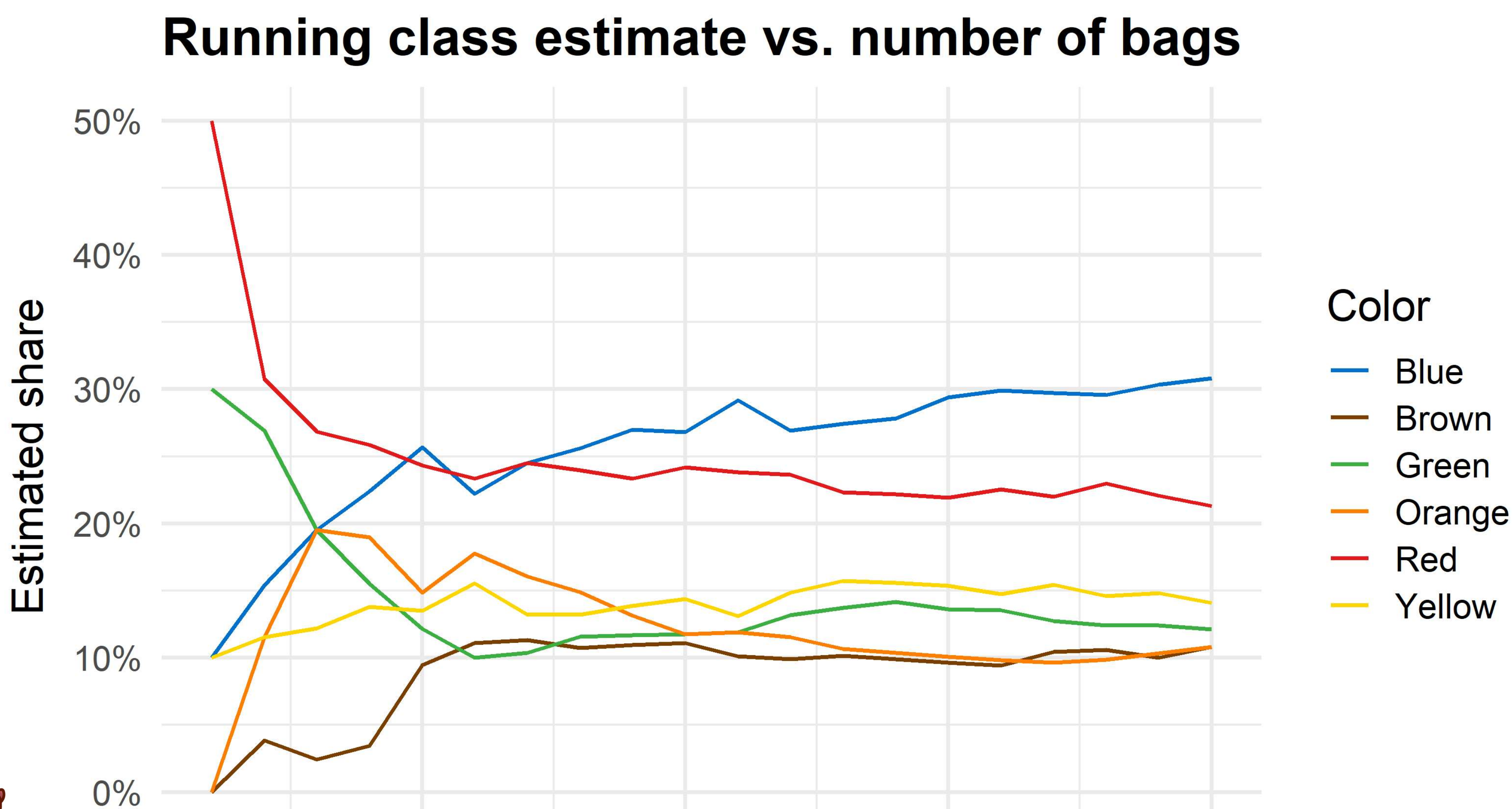
Sample.Size	Estimate.stability	Bag.to.bag.variation
Individual	Low	High
Paired	Medium	Medium
Class-wide	High	Low

- One bag can look quite different from another.
- Pooling more bags smooths out that variation.
- If we doubled the number of bags again, what would you expect to see in the pooled chart?


Overall M&M Color Distribution (pie)
300
0
100
200
x
Count
Color
Blue
Brown
Green
Orange
Red
Yellow


Overall M&M Color Distribution (bar)
Total Count
0
25
50
75
Blue
Brown
Green
Orange
Red
Yellow

How our estimate stabilizes



Advanced Sampling Concepts

Relating to Sampling Methods

- Simple random sampling
  - Each M&M package as a random sample
- Stratified sampling
  - If we sorted M&M bags by production date,
  - Could this improve representativeness?
- Cluster sampling
  - If we sampled entire boxes of M&M packages
  - Potential production batch effects?
- Systematic sampling
  - If we selected every nth M&M package from production line
  - Could introduce cyclical biases?

Potential Biases in M&M Sampling

- Production process biases
  - Color distribution variations between factories
  - Akin to sampling frame bias in surveys
- Selection bias
  - If students choose their favorite color of package
  - Akin to non-random sample selection in research
- Measurement bias
  - Errors in counting or recording M&M colors
  - Akin to survey response errors
- Non-response bias
  - If some students don't participate or eat their M&M's
  - Akin to survey non-respondents

Importance of Representative Samples

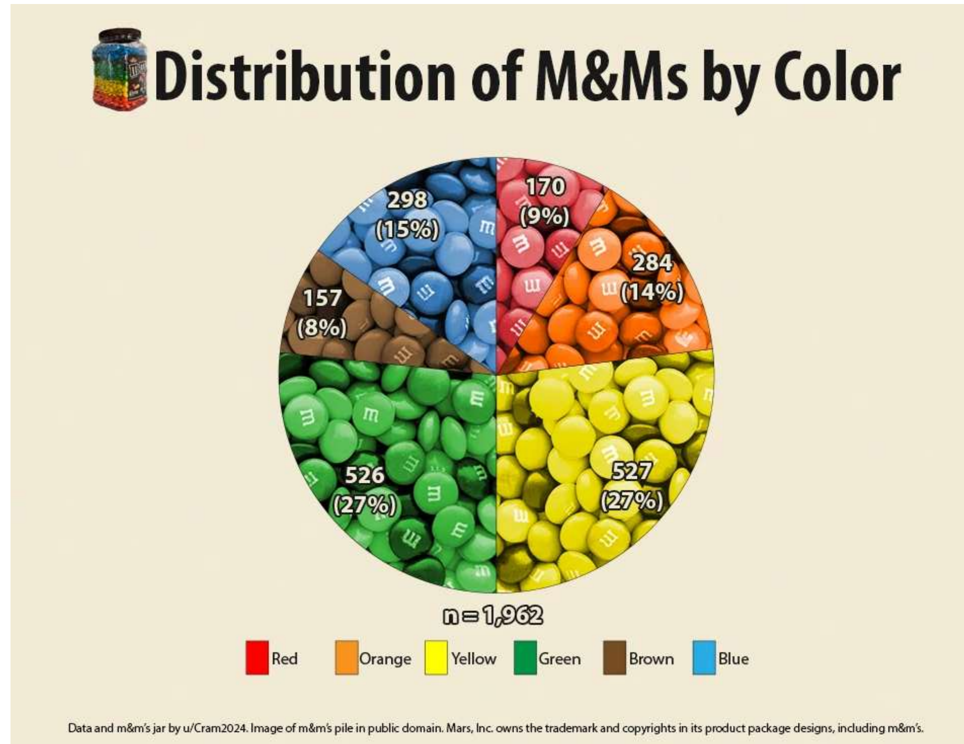
- What if we only sampled from one factory?
- Implications for psychological research
  - Generalizing from sample to population
  - External validity of research findings

- Strategies for improving representativeness
  - Increasing sample size (more M&M packages)
  - Diversifying sample sources (different stores, batches)
  - Random selection procedures
  - Weighting techniques for unequal probability samples

Wrapping Up...

Key Takeaways

1. Sampling lets us learn about populations with manageable data.
2. More data (more bags) leads to more stable estimates.
3. How we sample matters: source, selection, and recording affect results.
4. We can make smarter designs to improve generalization.



https://www.reddit.com/r/da